

Take-Home Assessment - AI-Collaborative Development

Build a full-stack application that consumes a public stock API and displays intraday market data.

You are encouraged to use AI assistants (Claude, Copilot, ChatGPT, etc.) - this reflects how we work. We will evaluate both the quality of the code you submit and how effectively you collaborate with AI. You own the final code - if AI generates something, you are expected to review, improve, and make it production-ready

Requirements

Backend

Using this public stock API (example uses TSLA):

```
1 curl "https://query1.finance.yahoo.com/v8/finance/chart/TSLA?
   interval=15m" -H "User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)
   AppleWebKit/537.36"
```

- Create a self-hosted C# .NET 8+ or Node.js/TypeScript API
- Expose an endpoint that:
 - Takes a stock symbol as a parameter
 - Queries intraday data from the last month
 - Groups results by day
 - Returns JSON in this format and precision:

```
1 [
2   {
3     "day": "2009-01-30",
4     "lowAverage": 40.2958,
5     "highAverage": 49.7534,
6     "volume": 49073348
7   }
8 ]
```

Frontend

- Build a UI (React, Angular, or Vue) that consumes your backend API
- Allow the user to enter a stock symbol and view the results
- Display the data in a meaningful way (table, chart, or both)
- Basic error handling for invalid symbols or failed requests

Deliverables

1. **Working repository** - commit(s) to a GitHub repository and provide the link (PREFERABLE), OR zip the contents and share
2. **README.md** - Setup and run instructions so a reviewer can build and run both backend and frontend locally
3. **PROMPT_LOG.md** - A log of the AI prompts you used during this exercise. For each entry include:
 - The prompt you sent
 - A brief note on why you chose that prompt (what were you trying to learn or achieve?)
 - What you kept, changed, or rejected from the AI output and why
4. Description and reasoning behind any manual changes made outside of AI

Expectations

- All requirements are met and the solution is fully functional
- Highest quality, suitable for a production deployment
 - Best practices, maintainable, SOLID, modern, etc.
- Assume this is an MVP with requirements expected to grow
- The prompt log should show thoughtful AI collaboration, not just copy-paste